Module 1

The Nature and Purpose of Intelligent Systems

Assessment Tasks

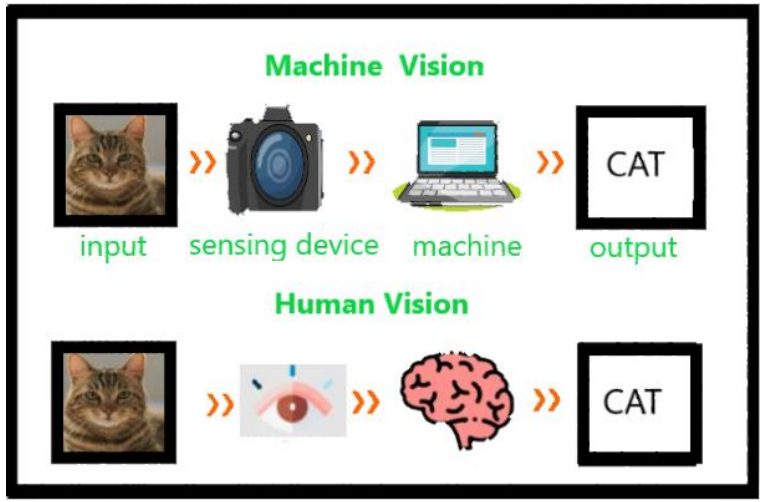
Assessment Tasks
12-week duration

No.	Type and Description	Assessment due	Weighting	Subject Learning Outcome(s) assessed
1	**Title:** Case Study **Delivery Type:** Individual **Outcome:** Written Report **Volume of assessment:** 1500 words (+/- 10%)	Module 5 (Week 5)	25%	a, b, d
2	**Title:** Programming Task **Delivery Type:** Individual **Outcome:** Program **Volume of assessment:** Code + Manual (500 words +/- 10%)	Module 8 (Week 8)	35%	a, c, e
3	**Title:** Final Project **Delivery Type:** Group & Individual **Outcome:** Presentation, Code and Individual contribution report **Volume of assessment:** 10-15 minutes total (group), Individual report (250 words +/- 10%) plus code	Module 12 (Week 12)	40%	a, b, c, d, e

Grading system

- You have only one (1) submission opportunity, there is no resubmission option. Late assessments result in 10% deduction per each day late. If there is no submission on the 5th day, it will result in a fail (0%).
- Grades in higher education banding:
  - 0–49% = F
  - 50–64%=P
  - 65–74%=C
  - 75–84%=D
  - 85–100%=HD

Machine Vision and Natural Language processing

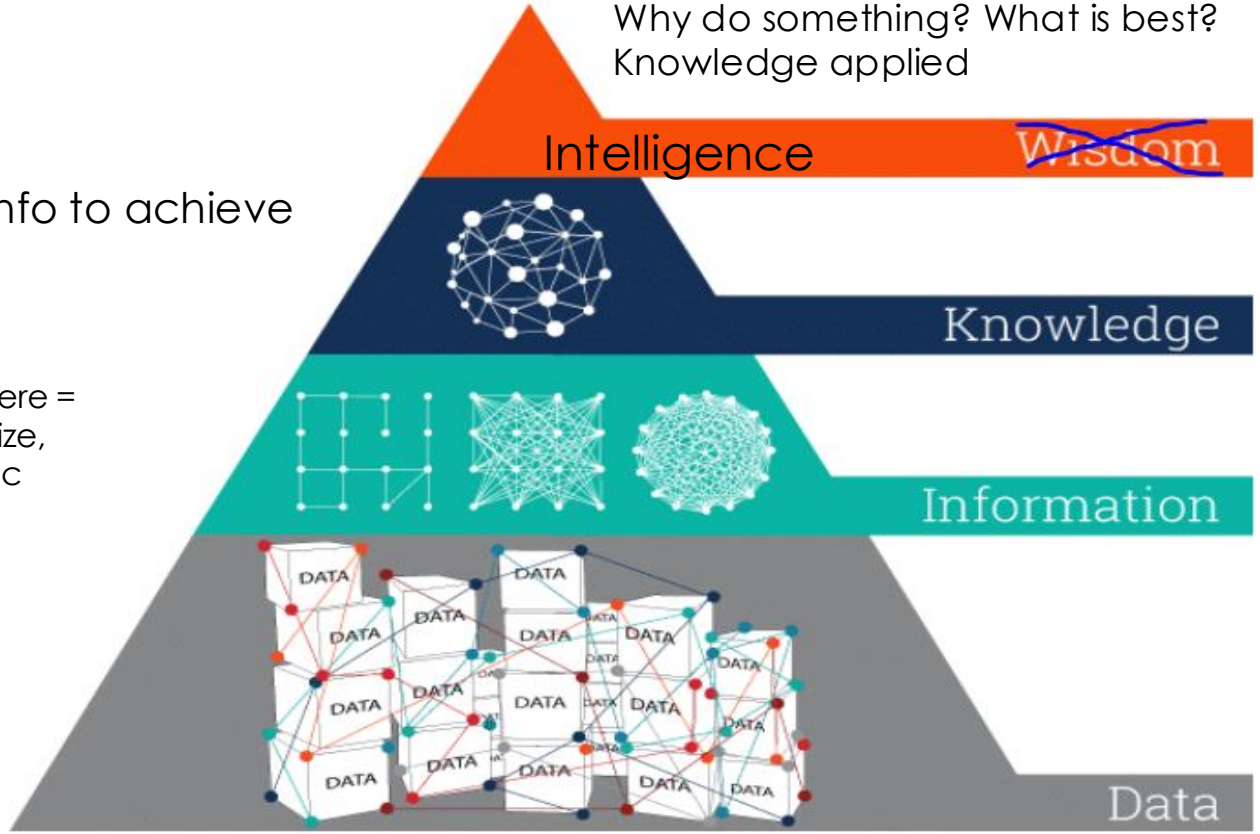


What is Data, Information, Knowledge and Intelligence ?

Why do something? What is best?
Knowledge applied

How? = apply info to achieve our goal

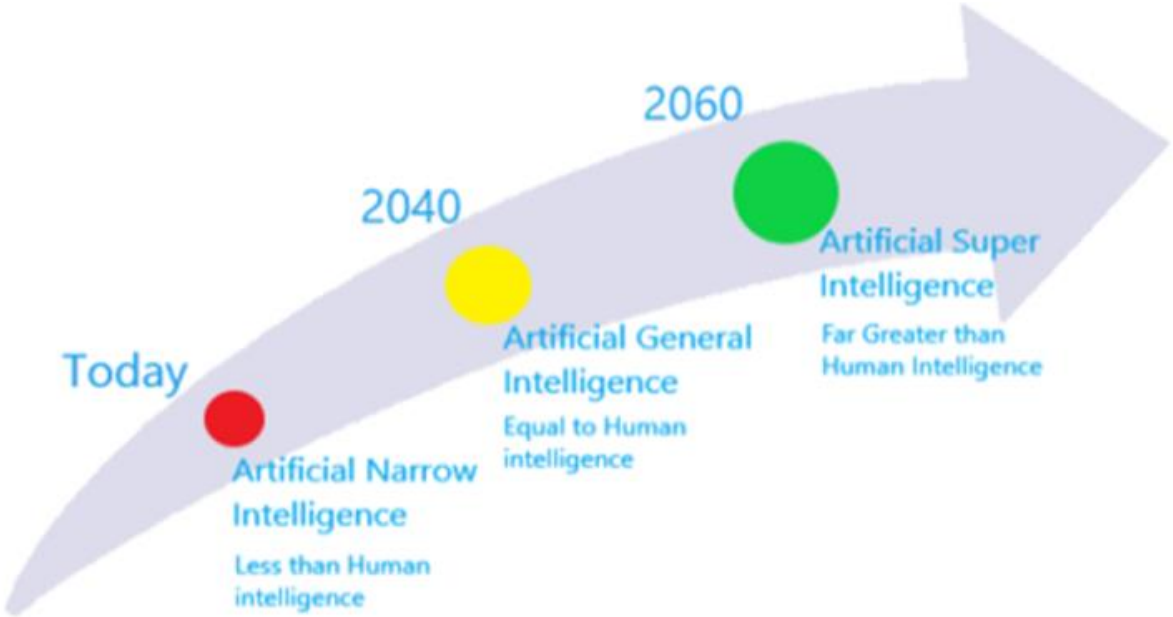
Who, What , When, Where = easy to measure, visualize, analyse data for specific purpose



Each step up the pyramid answers questions about and adds value to the initial data.

Collection of facts in a raw or unorganized form

Future Revolution of AI



Future Revolution of AI [3]

Types of Artificial Intelligence

	Artificial Narrow Intelligence (ANI)	Stage-1	Machine Learning	Specialises in one area and solves one problem
	Artificial General Intelligence (AGI)	Stage-2	Machine Intelligence	Refers to a computer that is as smart as a human across the board
	Artificial Super Intelligence (ASI)	Stage-3	Machine Consciousness	An intellect that is much smarter than the best human brains in pratically every field

Source: Great Learning | Infographic design by Antonio Grasso

Deltalogix

deltalogix.blog

Overview of Intelligent Systems

- Ever since the advent of computers, computer scientists have wondered if they might be made to learn

Mitchell, T. (1997), Machine Learning, McGraw-Hill

- Intelligent systems are computer-based systems that exhibit human-like intelligence, including the ability to perceive, reason, learn, and make decisions in complex environments.
- They leverage techniques from artificial intelligence (AI) to solve problems, adapt to changing conditions, and provide automated solutions
- Examples of Intelligent systems around you?

Overview of Intelligent Systems

- There are many definitions for what an Intelligent System is, a reasonable definition is as follows

- *An intelligent system operates in an environment with other agents, possesses cognitive abilities such as perception, action control, deliberative reasoning or language use, follows behavioural principles based on rationality and social norms, and has the capacity to adapt through learning.*

Molina, M. (2020). What is an intelligent system?. *arXiv preprint arXiv:2009.09083*.
https://www.researchgate.net/publication/344334868_What_is_an_intelligent_system

- So, what makes an Intelligent System Intelligent?

What is Intelligence Composed of?

- Reasoning
- Learning
- Problem Solving
- Perception
- Linguistic Intelligence

Human Intelligence	Machine/Artificial Intelligence
Percieve by patterns	Percieve by set of rules and data
Store and recall information by patterns	Store and recall information using search algorithms
Can figure out the complete object even if some parts are occluded/missing	Still not entirely possible

{ text and image processing, data integration, knowledge representation, machine learning, genetic programming}

Machine as intelligent agent shall have camera and microphones as sensors, telephone calls as actuators to perform classification and/or making decisions

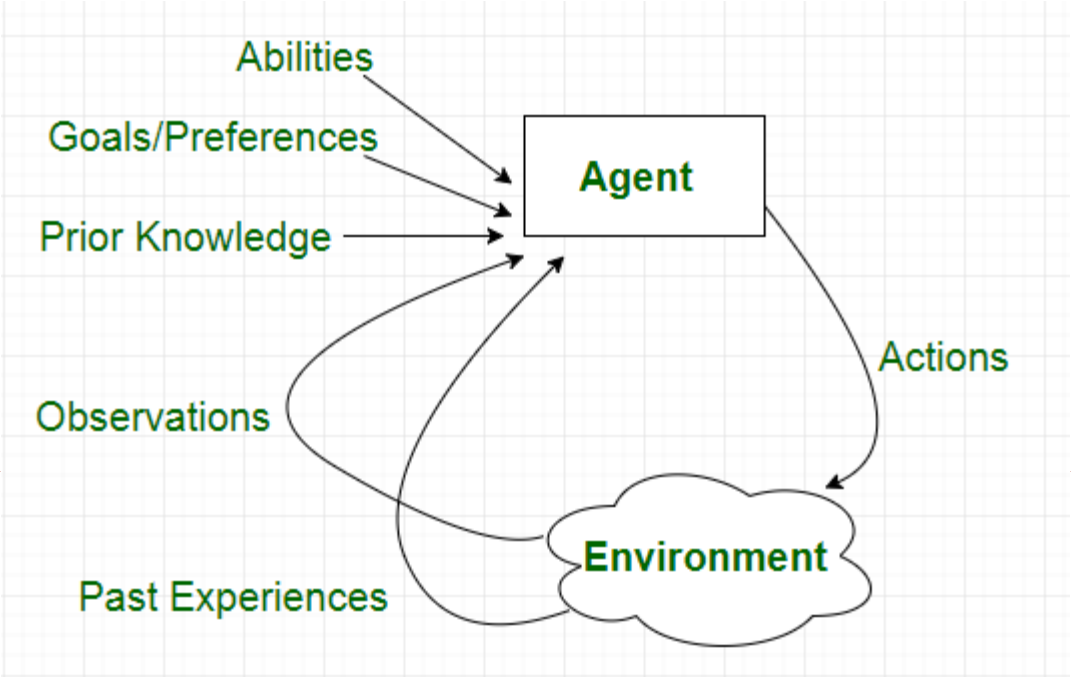
Overview of Intelligent Systems

- At the heart of an Intelligent System is an appropriate Machine Learning Algorithm What is it?

- What makes a Machine Learning Algorithm different than other algorithms is its ability to adapt (or 'learn') from experience my notes

- The precise characterization of an intelligent system is non-trivial because it is based on terms related to cognition, an area that is not fully understood and admits different interpretations
  - Some of the used terms can even change with the proposal of new computational models of intelligence and new scientific findings related to our understanding of the mind.

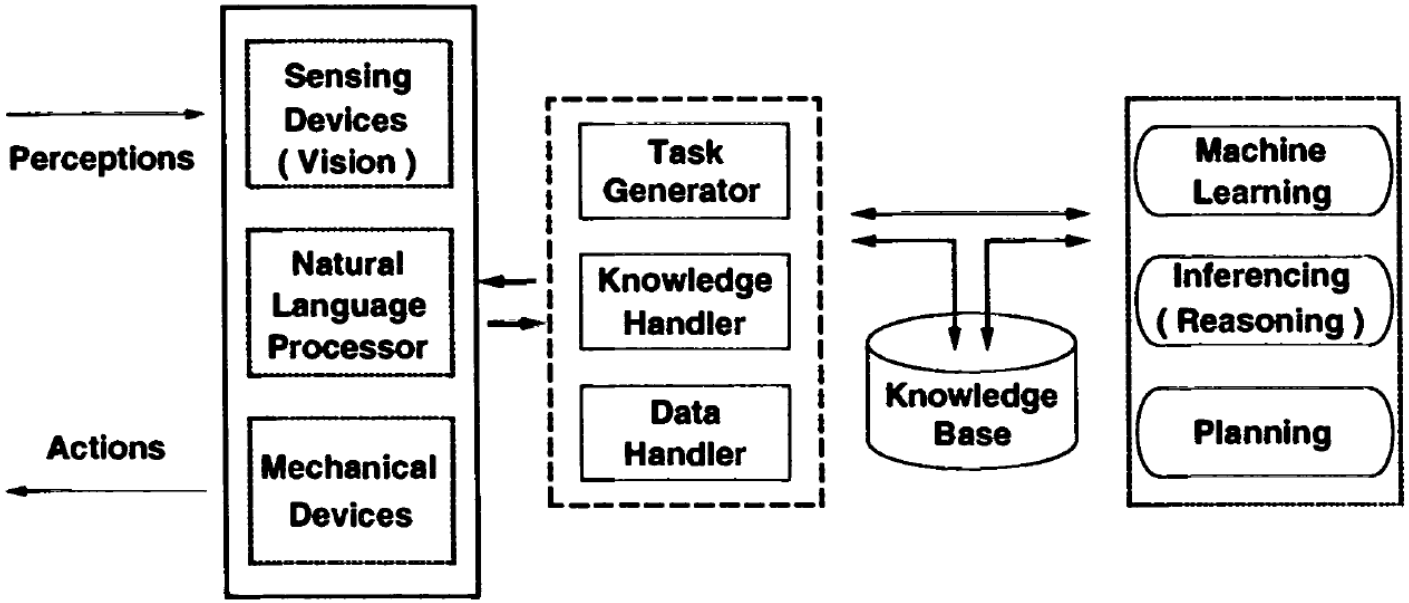
Intelligent software agents



A **software agent** is a computer program that **acts on behalf of a user or another program**, using **autonomy** to make decisions or perform tasks **without needing constant human guidance**.

Overview of Intelligent Systems

- Older schematic representation of an intelligent system

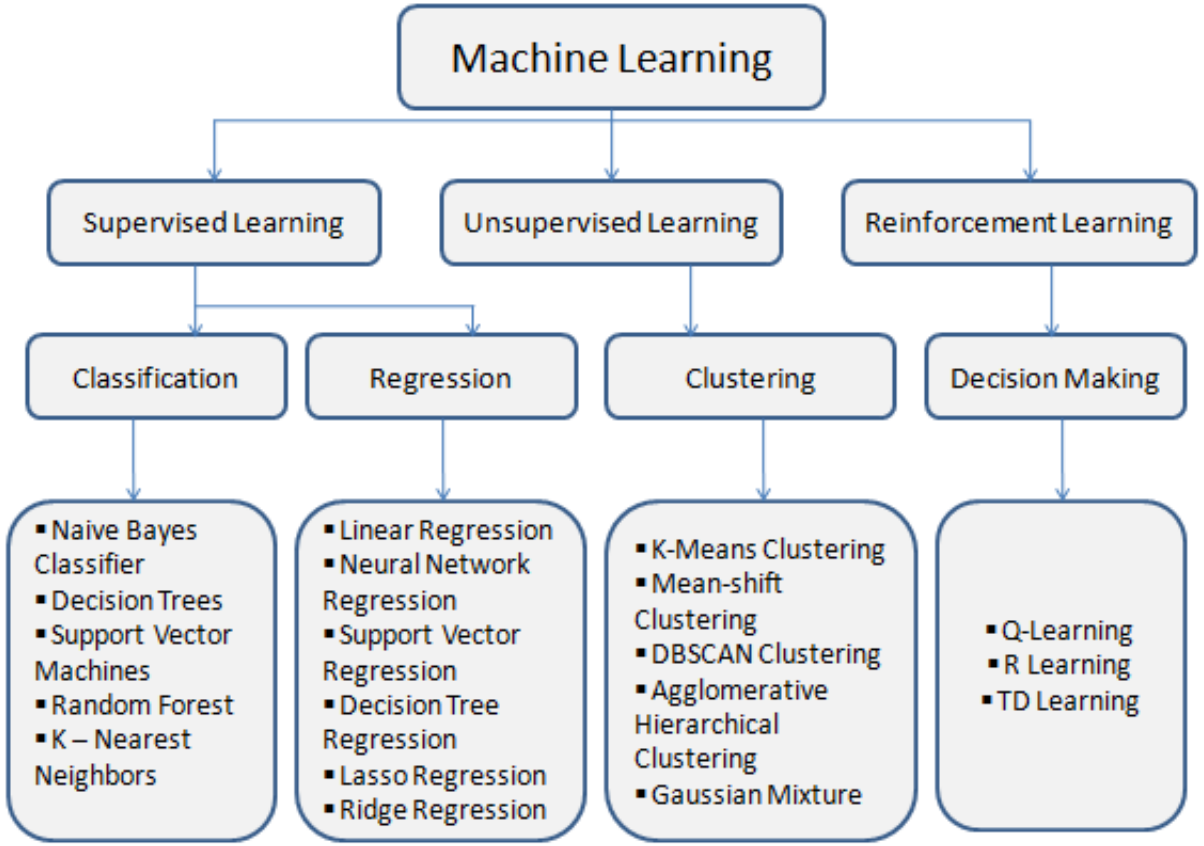


Jang, J.S., Sun, C.T., Mizutani, E., (1977), *Neuro-Fuzzy and Soft Computing*, Prentice-Hall

Overview of Intelligent Systems

- While we refer to these systems as being intelligent systems, we have to remember we are not talking about general AI, but a narrow form of AI
- The systems, while not being capable of human-like intelligence do exhibit intelligent-like behavior
  - Possessing the ability to learn
- There are many different types of Machine Learning algorithms but these fall into roughly 4 broad categories
  - Supervised
  - Unsupervised
  - Semi-supervised
  - Reinforcement learning

Machine Learning Algorithms



Overview of Intelligent Systems

- Supervised Learning
  - Supervised learning methods or algorithms include learning algorithms that take in data samples (known as training data) and associated outputs (known as labels or responses) with each data sample during the model training process
  - The main objective is to learn a mapping or association between input data samples x and their corresponding outputs y based on multiple training data instances
  - This learned knowledge can then be used in the future to predict an output y' for any new input data sample x' which was previously unknown or unseen during the model training process.

Overview of Intelligent Systems

- Supervised Learning
  - These methods are termed as supervised because the model learns on data samples where the desired output responses/labels are already known beforehand in the training phase.
  - Supervised learning methods are of two major classes based on the type of ML tasks they aim to solve
    - Classification
    - Regression

Overview of Intelligent Systems

- Supervised Learning
  - A supervised algorithm is shown the "right answer" for a set of sample data and finds a function which approximates the relationship between the inputs and outputs
  - This functional mapping takes the general form y = f(x) — specify your target output y, provide your inputs x, and the ML algorithm will learn the optimal f() by finding patterns in the data

y = f(x)

	Description	Training Phase	Live Model
y	Output	Supplied	Predicted
x	Input	Supplied	Supplied
f()	Functional mapping	Learned	Used to generate predictions

Overview of Intelligent Systems

- **Popular supervised learning algorithms**

Regression:

- Linear regression
- Random forest
- Multi-layer perceptron
- Convolutional deep neural networks

Classification:

- Logistic regression
- Support vector machines
- Convolutional deep neural networks
- Naive Bayes

Overview of Intelligent Systems

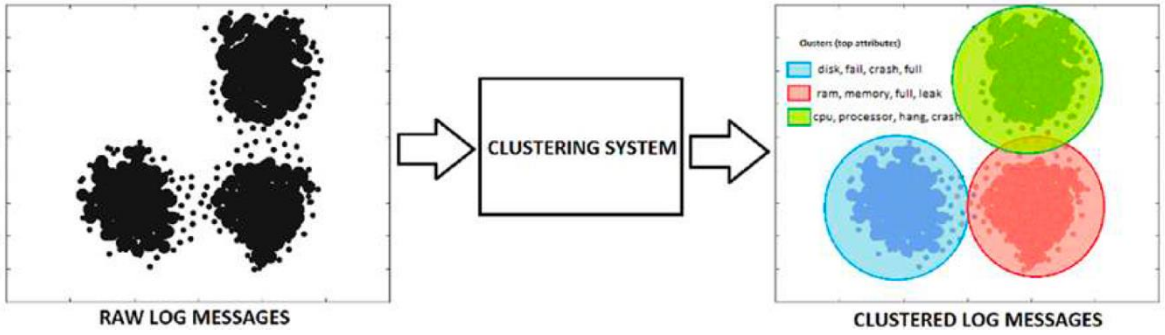
- Unsupervised Learning
  - In unsupervised learning, the machine is not provided labeled examples or previous patterns on which to base the analysis of the data inputs
  - The machine must uncover patterns and draw inferences by itself, without having the correct answers
  - It will classify or cluster data by discovering the similarity of features on its own
  - Using unsupervised learning, the machine would be fed millions of pictures of dogs, without labeling them as dogs
  - There is more uncertainty in the results of unsupervised learning
    - Can gain a lot of information from these models that was previously unavailable to view just by looking at the raw data.

Overview of Intelligent Systems

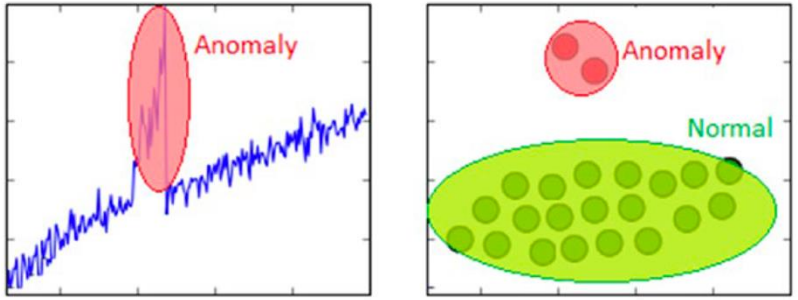
- Unsupervised Learning
  - Unsupervised learning methods can be categorized under the following broad areas of ML tasks relevant to unsupervised learning.
    - Clustering
    - Dimensionality reduction
    - Anomaly detection
    - Association rule-mining

Overview of Intelligent Systems

- Unsupervised Learning
  - examples
    - Clustering



    - Anomoly Detection



Overview of Intelligent Systems

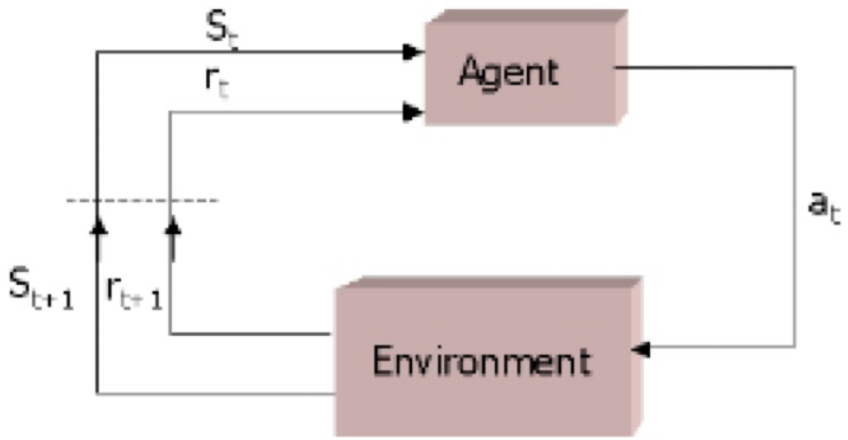
- Semi-Supervised Learning
  - The semi-supervised learning methods typically fall between supervised and unsupervised learning methods
  - These methods usually use a lot of training data that's unlabeled (forming the unsupervised learning component) and a small amount of pre-labeled and annotated data (forming the supervised learning component)
  - Popular reinforcement learning algorithms:
    - PU classification
    - Transductive SVM
    - Co-training

Overview of Intelligent Systems

- Reinforcement Learning
  - The reinforcement learning methods are a bit different from conventional supervised or unsupervised Methods
  - We have an agent that we want to train over a period of time to interact with a specific environment and improve its performance over a period of time with regard to the type of actions it performs on the environment
  - Typically the agent starts with a set of strategies or policies for interacting with the environment
    - On observing the environment, it takes a particular action based on a rule or policy and by observing the current state of the environment
    - Based on the action, the agent gets a reward, which could be beneficial or detrimental in the form of a penalty
    - It updates its current policies and strategies if needed and this iterative process continues till it learns enough about its environment to get the desired rewards

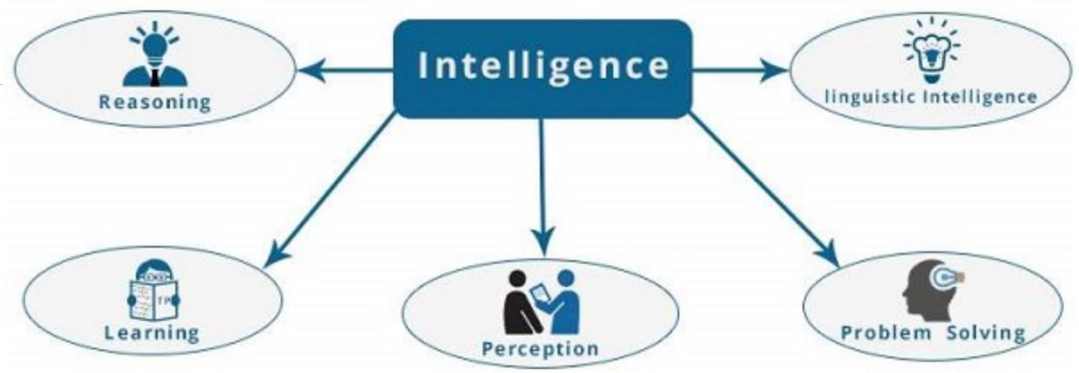
Overview of Intelligent Systems

- Reinforcement Learning
  - Some autonomous vehicles learn to drive through reinforcement. These cars are instructed to get from point A to point B under only two broad conditions: obey the rules of the road, and don't crash. The rest is learned through trial and error.
  - The traditional reinforcement-learning update cycle is shown below
    - at time *t*, an agent will sense it's current state S_t and choose action at a_t time *t* to maximize its reward *r*
    - This will result in a possibly new state S_{t+1} and time time *t+1*, and a corresponding reward r_{t+1}, which then feeds back into the agent so it can now choose a new action based on the new state.



Overview of Intelligent Systems

- What is Intelligence
  - The ability of a system to calculate, reason, perceive relationships and analogies, learn from experience, store and retrieve information from memory, solve problems, comprehend complex ideas, use natural language fluently, classify, generalize, and adapt new situations
  - Intelligence is intangible - composed of



Source: Tutorials Point

[https://www.tutorialspoint.com/artificial_intelligence/artificial_intelligent_systems.htm]

Overview of Intelligent Systems

- Difference between Human and Machine Intelligence
  - Humans perceive by patterns whereas the machines perceive by set of rules and data
  - Humans store and recall information by patterns, machines do it by searching algorithms. For example, the number 40404040 is easy to remember, store, and recall as its pattern is simple
  - Humans can figure out the complete object even if some part of it is missing or distorted; whereas the machines cannot do it correctly

Overview of Intelligent Systems

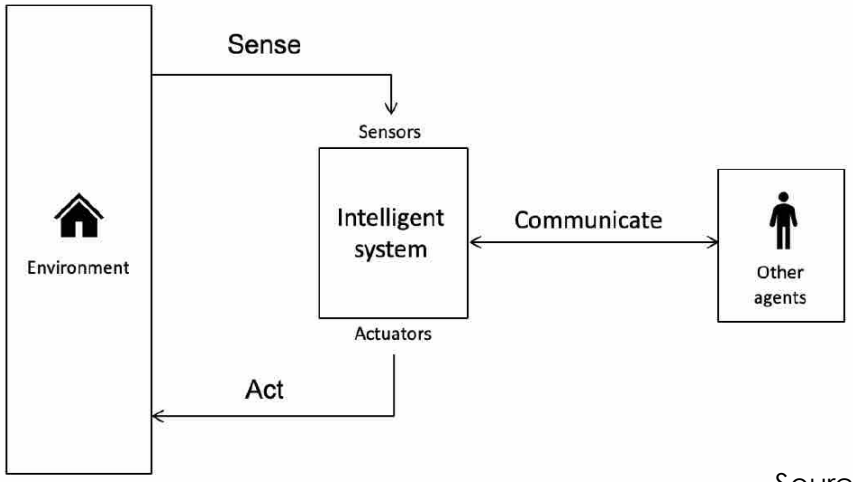
- Our original definition of an Intelligent System indicates four properties that these systems possess which can be mapped to the elements of intelligence on the previous slide

  1. Working in an environment with other agents
  2. Having primary cognitive abilities
  3. Following principles about rationality and social norms
  4. Capacity to adapt

- Each of these presents additional concepts and categories in which the elements of a system may be classified

Overview of Intelligent Systems

- 1 Working in an environment with other agents
  - The system observes features from the environment through sensors and performs actions using actuators
  - The use of sensors and actuators (real or virtual) separates the body of the intelligent system from the rest of the environment. This characteristic is called embodiment.

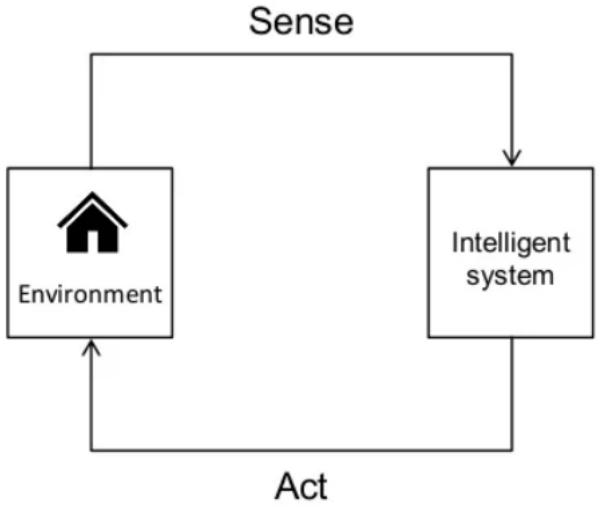


Source: Molina, M. (2020). What is an intelligent system

Overview of Intelligent Systems

- Environment
  - The degree of complexity of the environment with respect to the agent can be described with the following features

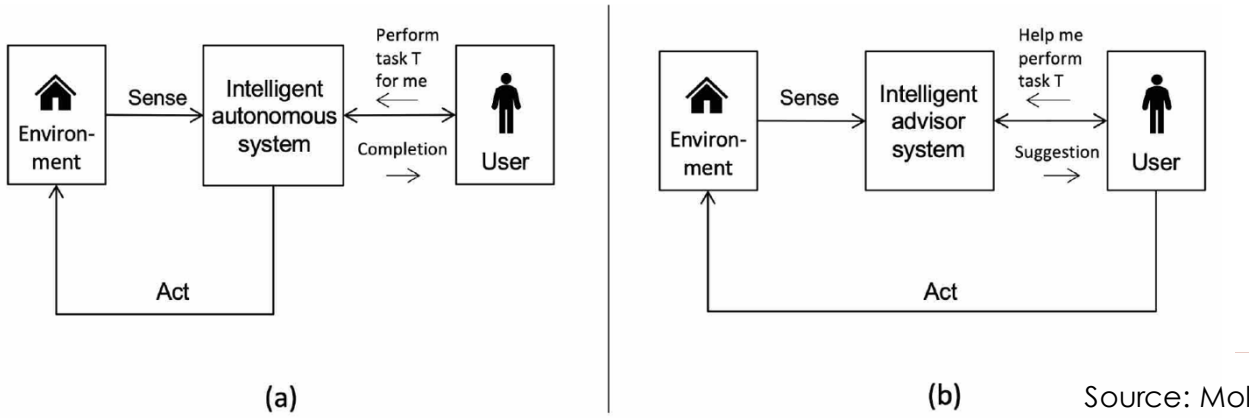
- **Static / dynamic**
  The environment (doesn't change / changes) while an agent is deliberating
- **Discrete / continuous**
  The state of the environment, time, percepts or actions (are discrete / are continuous)
- **Fully-observable / partial-observable**
  Sensors (detect / don't detect) all aspects that are relevant to the choice of action
- **Deterministic / stochastic**
  The next state of the environment (is / isn't) completely determined by the current state and the action
- **Episodic / sequential**
  Actions (don't influence / influence) future actions
- **Known / unknown**
  The outcomes for actions (are known / are not known) by the agent in advance



Russell, S., Norvig P. (2009). Artificial Intelligence: A Modern Approach (3rd edition). Pearsons Education Limited.

Overview of Intelligent Systems

- Other Agents
  - An intelligent system may interact with other agents as part of an organization

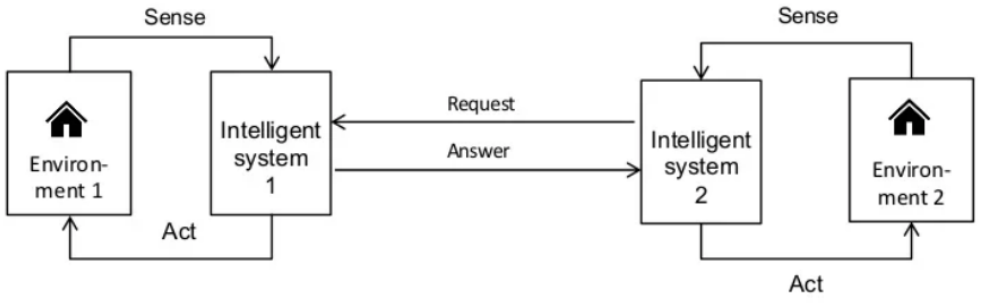


Source: Molina, M. (2020). What is an intelligent system

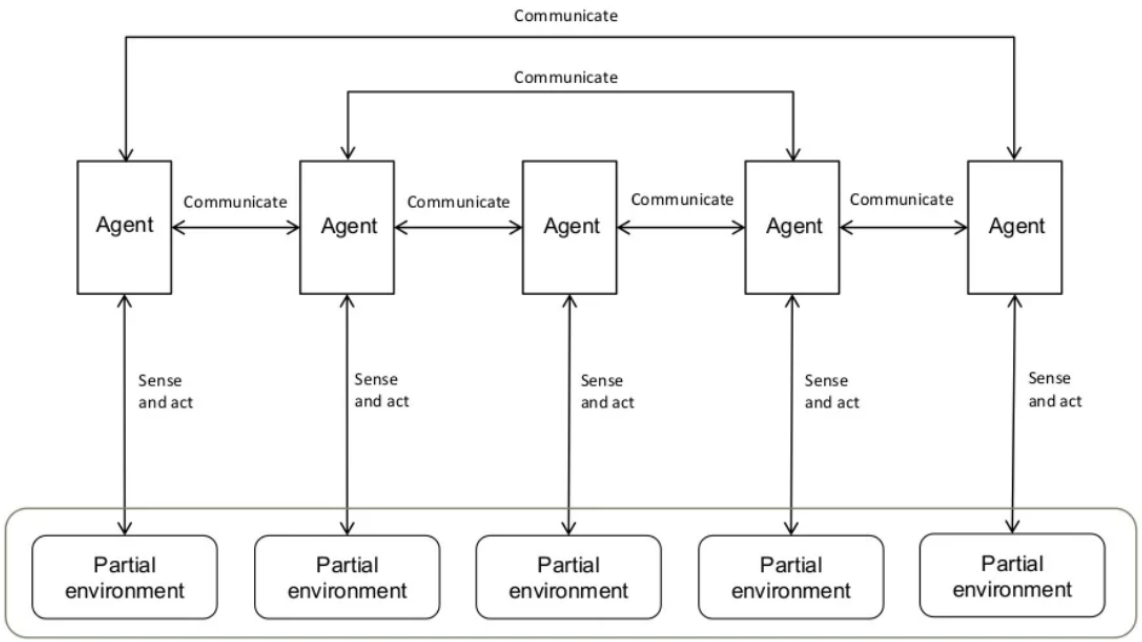
- (a): An autonomous system acts in the environment to help the user
- (b): An advisor system helps the user act in the environment

Overview of Intelligent Systems

- An Intelligent may interact with another system



or be part or a larger multi-agent system



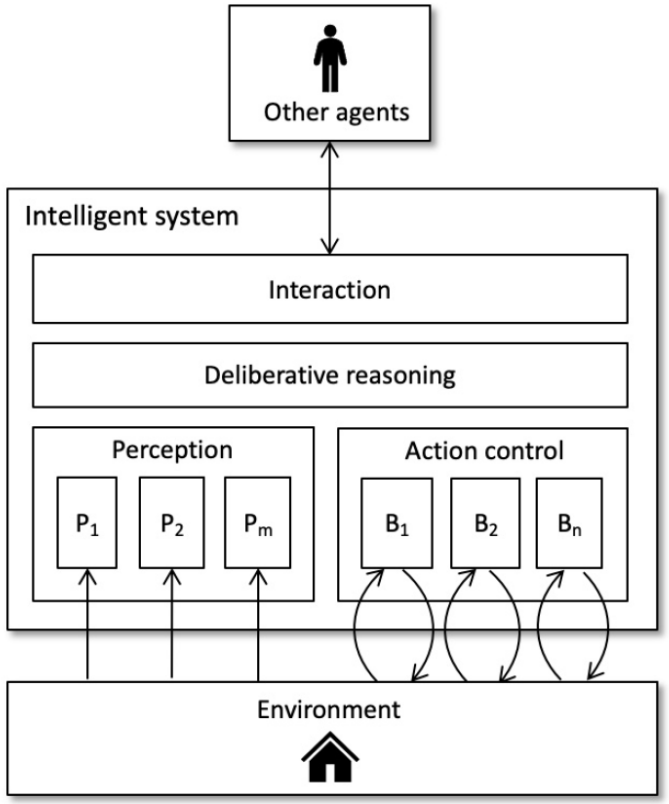
Source: Molina, M. (2020). What is an intelligent system

Overview of Intelligent Systems

- 2 Having primary cognitive abilities
  - We can classify cognitive abilities in roughly 4 categories which are usually present in intelligent systems
    - Perception - Perception may be understood as the interpretation of sensory information to form a mental representation of the world
    - Action control - Action control refers to the ability of an agent to control the execution of its own actions
    - Interaction - The ability to interact with other agents is based on using a common language
    - Deliberative reasoning. An intelligent agent may be able to reason in order to plan actions or diagnose problems

Overview of Intelligent Systems

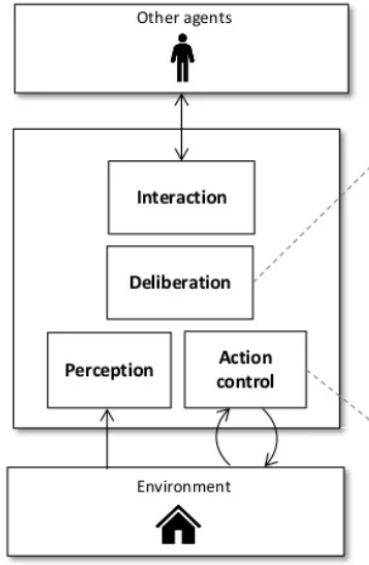
- Perception and action control may be divided into multiple separate components
- Deliberation and communication usually operate sequentially which is necessary for coherent reasoning
- The global behavior that emerges from individual behavior controllers is reactive in the sense that it is able to react quickly to events of the environment
- Using symbols has been been pointed out as a relevant feature of intelligence
- Symbols are present in the language used for agent interaction
- An agent may establish the meaning of symbols through perception and action control



Source: Molina, M. (2020). What is an intelligent system

Overview of Intelligent Systems

- Must achieve balance between deliberation and reaction



Deliberation

Making decisions about what to do based on justifiable reasons

Advantages:

- Reactive behaviors can be inhibited to reach more useful long-term goals
- Decisions are consistent with own knowledge

Reactive behavior

Generation of instantaneous actions in response to a stimulus (e.g., animal reflexes or in decisions based on intuitions).

Advantage:

- Efficient reaction to dynamic events in a dynamic environment (it uses limited memory about the world)

Overview of Intelligent Systems

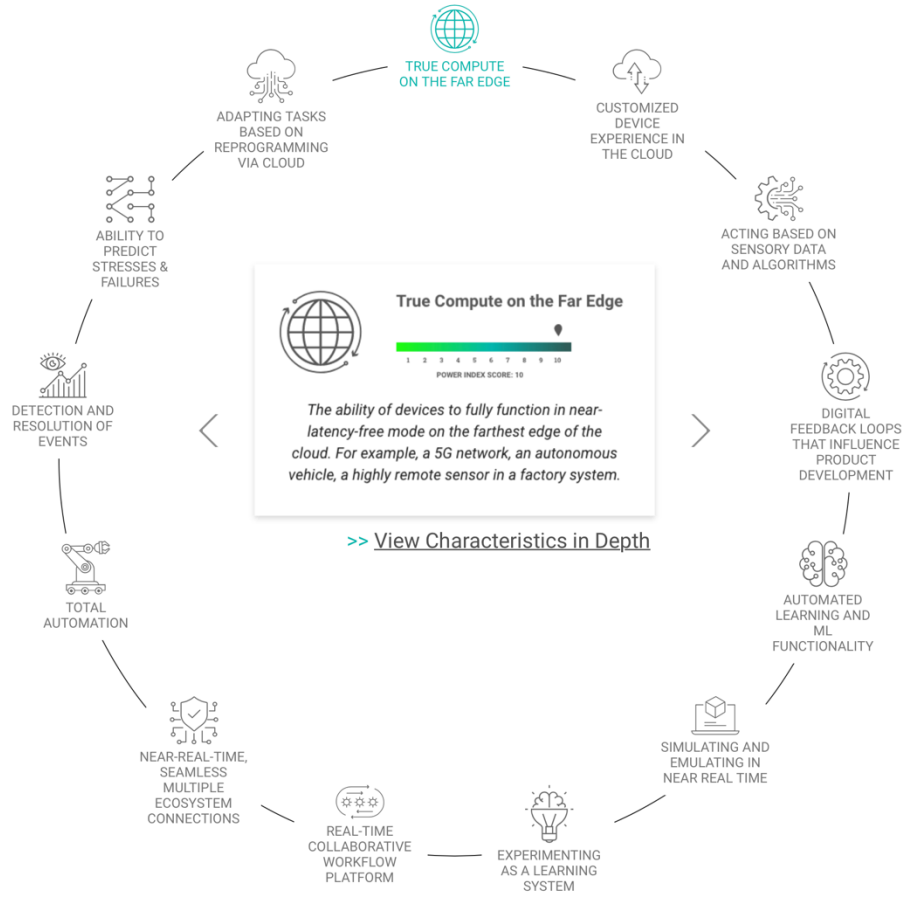
- 3 Following principles about rationality and social norms
  - An intelligent system is expected to follow two basic principles of behavior
    - The first principle is acting rationally and affects the individual behavior of the system with respect to its goals and the environment
      - An agent acts rationally if the decisions it makes about its actions seek to maximize a measure of performance
      - Acting rational is closely related to the notion of intelligence
    - The second principle is about following social norms and affects the behavior of the system with respect to other agents
      - an intelligent system trained using biased data may be not able to ensure a fair operation
      - It is important to ensure that the system operates in accordance with ethical values such as ensuring fairness (avoiding group biases) or limiting harmful use

Overview of Intelligent Systems

- 4 Capacity to adapt
  - The capacity for adaptation allows an intelligent system to operate effectively in complex dynamic environments
    - a system is said to be self-adaptive if it is able to modify the way it performs a certain task in response to changes in the environment so that the system can perform such task more efficiently
  - A first approach to achieve self-adaptation is by a dynamic re-combination of behaviors
  - A second approach for adaptation is through learning
    - Learning from experience
    - Learning from other Agents

Overview of Intelligent Systems

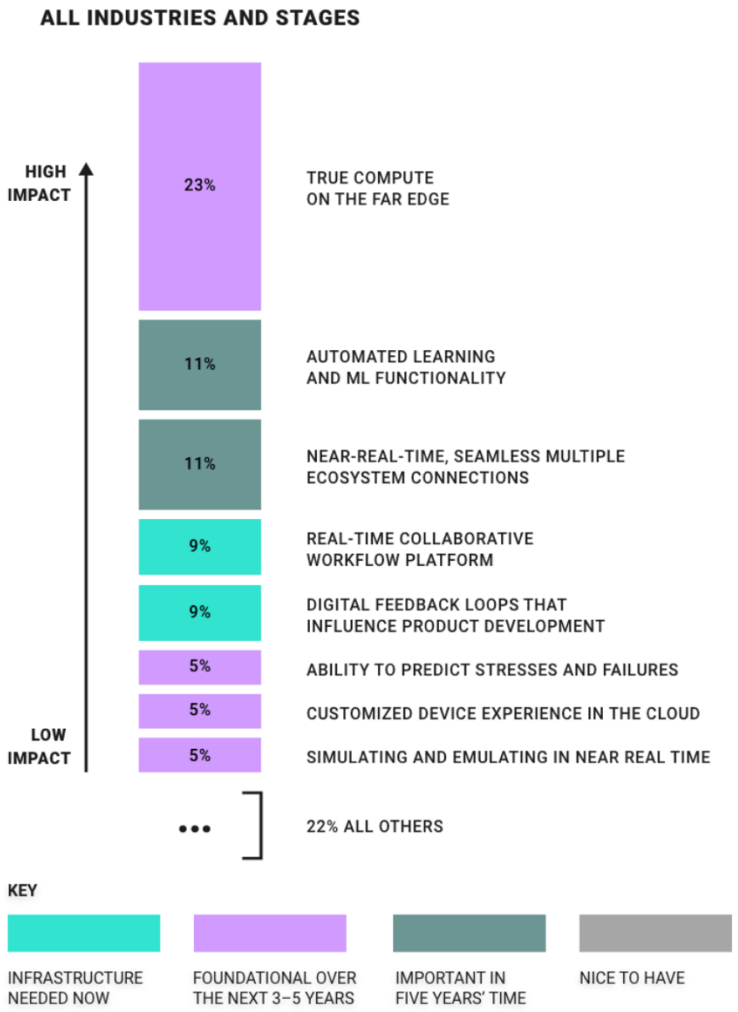
- Characteristics of Intelligent Systems (from a business perspective)



Source: Windriver
https://www.windriver.com/intelligent-systems

Overview of Intelligent Systems

- Impact on business



Source: Characteristics of an Intelligent Systems Future, Forbes, 2021

Overview of Intelligent Systems

- Examples of AI Applications
  - There are many examples of applications incorporating AI techniques
    - Automated Customer Support
    - Personalized Shopping Experience
    - Healthcare
    - Finance
    - Smart cars and drones
    - Travel and navigation
    - Social media
    - Smart home devices
    - Creative arts
    - Security and surveillance

Source: https://becominghuman.ai/10-powerful-examples-of-ai-applications-553f7f062d9f

Ethics in AI

- With the rapid development of artificial intelligence have come concerns about how machines will make moral decisions, and the major challenge of quantifying societal expectations about the **ethical principles that should guide machine behaviour**.
- To address this challenge, we(MIT) deployed the Moral Machine, an online experimental platform designed to explore the moral dilemmas faced by autonomous vehicles (A platform for gathering a human perspective on moral decisions made by machine intelligence, such as self-driving cars). https://www.moralmachine.net/
- This platform gathered **40 million decisions in ten languages from millions of people in 233 countries and territories**.
- Moral machine to build an inference engine in autonomous vehicles.

Ethics in AI contd…

- Result Analysis:
  - summarize global moral preferences.
  - document individual variations in preferences, based on respondents' demographics.
  - report cross-cultural ethical variation, and uncover three major clusters of countries.
  - show that these differences correlate with modern institutions and deep cultural traits.

- Discuss how these preferences can contribute to developing global, socially acceptable principles for machine ethics. All data used in this article are publicly available

Responsible AI in Practice

- https://twimlai.com/podcast/twimlai/responsible-ai-practice-sarah-bird/
- Listen the video with more emphasis from 20 to 30 min were it discusses on fairness and differential privacy.
- Post keywords on Discussion Forums.

- Activity 2:

Using the techniques outlined in 'How to read a paper' (Keshav, 2007), write a summary of the two following journal articles. The summaries should be posted to your private journal for Module 1, be no more than 200 words each (400 total) and should address:

Why solving that specific problem was important;

What intelligent system or AI technique was used; and

What were the ethics discussed in the work.

Weekly activity

- Consider the following situation:
- A bank is using an intelligent system to grant loans. The bank gives each customer a form to complete. In the form, one question asks the customers to give reasons as to why they will not be late in repaying the loan instalments. The system uses computer vision to read the hand-writing of the customer. It also uses natural language processing to sense the feeling of their writing (e.g., if it is polite, friendly, formal, etc.). After a year in which 2,000 loan forms were collected, the bank trains a ML algorithm to grant loans only to those customers whose writing shares a similar feeling, as such customers are never late in their loan repayments.
- Using the resources you have read in this Module, discuss the ethics of the situation given above. Ensure you also discuss the concept of fairness in the scenario.
- Describe each of these (ethics and fairness) in no more than 100 words.
- Post your answer to the Module 1 discussion forum.

The End